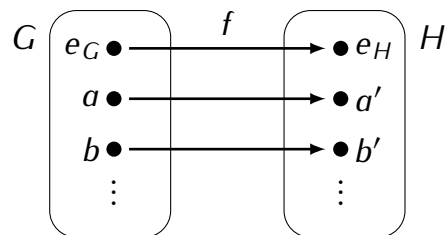


**Definition 14.1**

An *isomorphism of groups* is a group homomorphism which is both onto and 1-1.

**Theorem 14.2**

If  $g: G \rightarrow H$  is an isomorphism then the inverse function  $f^{-1}: H \rightarrow G$  is also an isomorphism.

**Theorem 14.3**

A homomorphism of groups  $f: G \rightarrow H$  is an isomorphism if and only if  $\text{Im}(f) = H$  and  $\text{Ker}(f) = \{e\}$ .

#### Definition 14.4

We say the group  $G$  is *isomorphic* to a group  $H$  if there exists an isomorphism  $f: G \rightarrow H$ . Then we write  $G \cong H$ .

#### Theorem 14.5

Isomorphism of groups is an equivalence relation:

- 1) For any group  $G$  we have  $G \cong G$ .
- 2) If  $G, H$  are groups such that  $G \cong H$  then  $H \cong G$ .
- 3) If  $G, H, K$  are groups such that  $G \cong H$  and  $H \cong K$ , then  $G \cong K$ .

Examples.

### Theorem 14.6 (Cayley's Theorem)

Let  $G$  be a finite group of order  $n$ . Then  $G$  is isomorphic to a subgroup of the symmetric group  $S_n$ .

**Definition 14.7**

An *automorphism* of a group  $G$  is an isomorphism  $f: G \rightarrow G$ .

**Definition 14.8**

Let  $G$  be a group. The *group of automorphisms* of  $G$  is the group  $\text{Aut}(G)$  whose elements are automorphisms of  $G$  and the group operation is given by composition of automorphisms.